

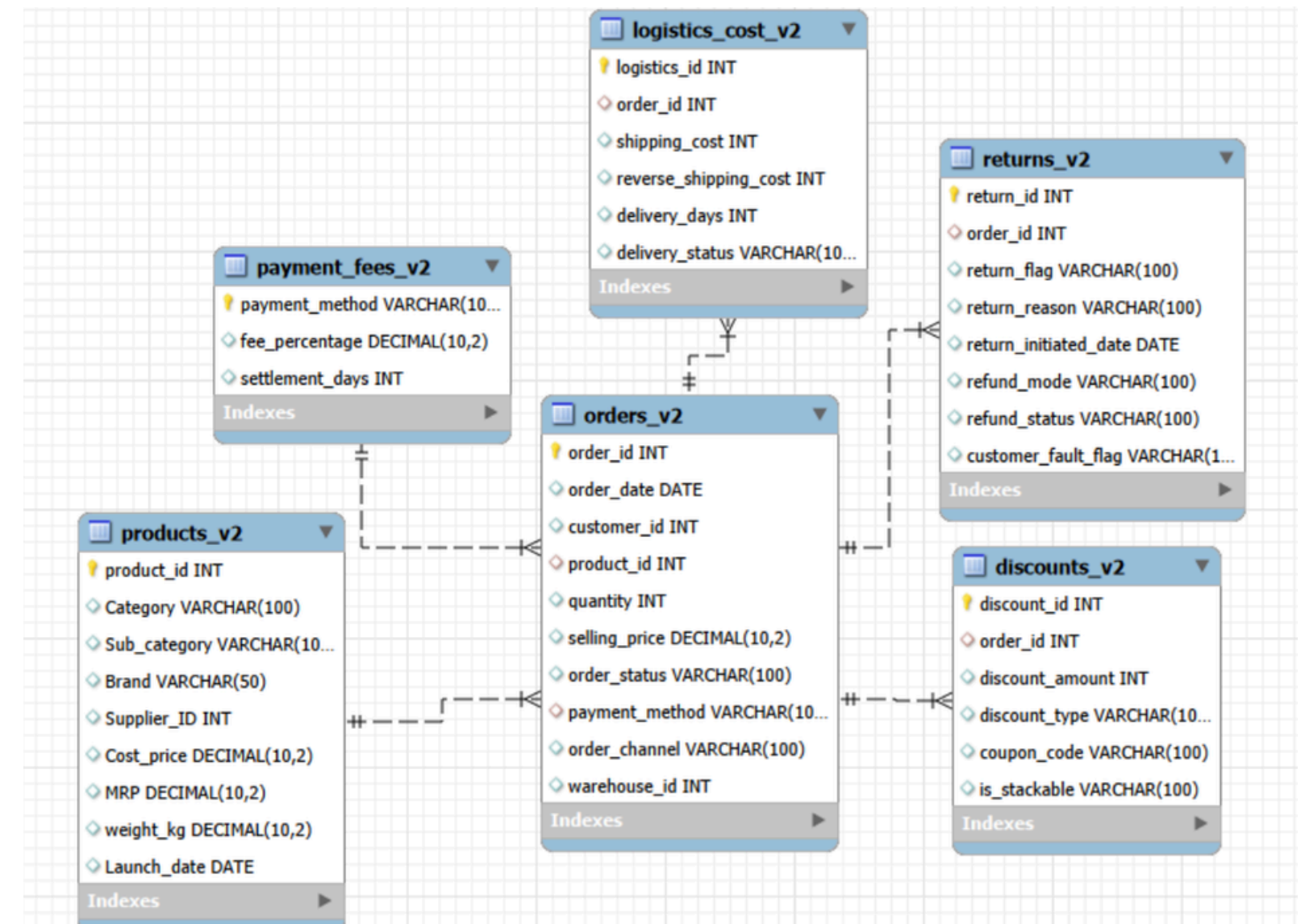
MARKETPLACE REVENUE LEAKAGE & GAP ANALYSIS USING SQL



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ENTITY-RELATIONSHIP DIAGRAM

- Schema follows a **Star schema** design — ideal for marketplace analytics
- The table **orders_v2** acts as the central hub connecting all 5 other tables
- **One-to-many relationships** exist between orders and related tables such as products, discounts, logistics, returns, and payment methods.



REVENUE VS PROFIT REALITY

INSIGHTS

- The Marketplace generated a Total revenue of **19.31million**. The total product procurement cost was **8.73 million**. The business achieved a gross profit of **10.58 million**
- Business retains more than half of its revenue as gross profit
- Revenue calculated only for Delivered and Shipped orders — ensuring accuracy
- This is Gross profit — does not yet account for discounts, logistics, payment fees, and returns

Total_Revenue	Total_Cost	Total_Profit
19308912.00	8725866.00	10583046.00

CATEGORY-WISE SALES & PROFIT

INSIGHTS

- **Electronics** has the **highest revenue** but a relatively lower margin — high volume, moderate efficiency
- **Grocery and Home** have **best profit margins** despite lower revenue — most efficient categories
- All categories show **similar margins** — indicating a uniform pricing strategy
- UNKNOWN category **₹26L revenue** — proper categorization could redistribute this across existing categories

category	Total_Revenue	Total_Cost	Total_Profit	Profit_Margin_Percentage
Electronics	3822772.00	1753860.00	2068912.00	54.120727
Fashion	3385655.00	1565841.00	1819814.00	53.750722
Home	3233382.00	1430228.00	1803154.00	55.766810
Grocery	3110732.00	1360626.00	1750106.00	56.260263
Beauty	3156721.00	1462032.00	1694689.00	53.685106
unknown	2599650.00	1153279.00	1446371.00	55.637143

LOSS-MAKING PRODUCTS

INSIGHTS

- Several SKUs generated negative net profit despite continued sales volume, indicating hidden profitability leakage.
- Loss-making products were driven by factors such as:
 1. **High costs**
 2. **Discounts**
 3. **Returns, and**
 4. **Operational expenses**

Recommendations:

The analysis highlights **opportunities for repricing, supplier renegotiation, or discontinuation of persistently unprofitable products.**

product_id	category	sub_category	brand	total_quantity	Total_Net_Profit
152	Electronics	SubCat-7	BrandC	14	-13042.00
23	Electronics	SubCat-10	BrandD	18	-11524.00
652	Beauty	SubCat-3	BrandB	26	-10448.00
871	Grocery	SubCat-6	BrandC	11	-9575.00
773	Home	SubCat-4	UNKNOWN	17	-8631.00
891	Home	SubCat-2	BrandB	29	-8624.00
1661	Grocery	SubCat-4	UNKNOWN	36	-8603.00
1374	Grocery	SubCat-5	BrandC	9	-8084.00
87	Home	SubCat-2	BrandB	8	-7605.00
278	Grocery	SubCat-1	BrandA	7	-7268.00
629	Fashion	SubCat-4	BrandC	10	-7148.00
789	Home	SubCat-8	BrandD	12	-6983.00
849	Electronics	SubCat-8	BrandC	19	-6343.00

DISCOUNT USAGE OVERVIEW

INSIGHTS

- **1 in every 3 orders (33.51%)** receives a discount, and **₹1,654,459** total revenue (**8.57% of total revenue**) is given away as discounts
- This suggests business is dependent on promotions to drive sales

Recommendations:

- Discounts could be targeted only on low-performing categories instead of blanket discounting.
- Every 1% reduction in discount rate saves approximately ₹193,000 in revenue"

Total_Discounted_orders	total_discount_amount
2163	1654459

PAYMENT METHOD POPULARITY

INSIGHTS

- Orders and revenue are **almost equally distributed** across all payment methods
- No single payment method dominates — indicating diverse customer payment preferences

Recommendations:

- **UPI** should be promoted as it has the lowest fee (1.2%) among known methods.
- UNKNOWN payment method has the highest orders (1,354) and revenue (₹4.17M), as it represents the major data quality concern

payment_method	Total_number_of_Orders	Total_sales_value
UNKNOWN	1354	4178191.00
UPI	1328	3968219.00
COD	1270	3880565.00
CARD	1257	3672135.00
WALLET	1245	3609802.00

DISCOUNT VS PROFIT GAP

INSIGHTS

- Discounting strategy is **ineffective** — more orders come without discounts anyway
- Discounts are **reducing profit without increasing volume**

Order_type	Total_orders	Average_Profit
Non-discounted_orders	4291	1611.560708
Discounted_orders	2163	930.827554

Recommendations:

- The current discounting strategy neither drives volume nor protects margins so immediate **discount optimization** is recommended

RETURN IMPACT ON REVENUE

INSIGHTS

- 156 returns out of 6,454 delivered/shipped orders, indicating returns are **not a major operational crisis**, but still a significant leakage source
- ₹520,032 revenue lost to **confirmed processed returns** — representing **2.69% of the total revenue of ₹19.3M**, directly reversed back to customers

Total_returned_orders	Total_Revenue_lost	Total_Profit_lost
156	520032.00	294899.00

Recommendations:

Addressing the root causes of those returned orders can directly recover a significant portion of the revenue loss

RETURN REASON ANALYSIS

INSIGHTS

- **Damaged products and wrong-item deliveries** caused the highest return-related revenue losses, highlighting major issues in **packaging, handling, and order fulfillment processes**.
- Returns were broadly distributed across all reasons, while late deliveries and unknown return reasons indicate **logistics inefficiencies and gaps in return data capture**.

return_reason	Count_of>Returns	Revenue_lost	Total_Profit_lost
Damaged	27	109810.00	58473.00
Wrong Item	27	100283.00	53533.00
UNKNOWN	27	88248.00	52991.00
Customer Changed Mind	27	81420.00	51193.00
Quality Issue	27	78574.00	41796.00
Late Delivery	24	72347.00	41800.00

LOGISTICS COST BURDEN

INSIGHTS

- Almost all orders in this data have values below ₹500 — confirms that low order value is the sole driver of logistics burden
- Multiple orders across different value ranges, all showing **65%+** logistics burden — **requires immediate policy intervention**

Recommendations:

- **Minimum order value of ₹500 recommended** — would eliminate all orders currently showing extreme logistics burden and significantly reduce overall leakage

order_id	Order_Value	total_logistics_cost	Logistics_Percentage
13798	215.00	271	126.05
14930	231.00	265	114.72
15910	218.00	209	95.87
4813	250.00	238	95.20
11884	354.00	324	91.53
13700	257.00	232	90.27
10193	329.00	296	89.97
21338	284.00	245	86.27
21716	220.00	176	80.00
8851	205.00	154	75.12
13218	290.00	194	66.90
18799	416.00	276	66.35
19550	238.00	156	65.55

PAYMENT FEE LEAKAGE

INSIGHTS

- **COD** generated the highest payment gateway fees, making it the **most expensive payment method** and significantly reducing profit margins.
- **UPI** was the **most cost-efficient payment option**, while unknown payment methods created gaps in accurate fee leakage tracking.

payment_method	Total_payment_gateway_fee	Total_Net_Profit	Total_Net_Profit_including_fee_percentage
COD	135819.78	2157485.00	2021665.23
CARD	77114.84	2025126.00	1948011.17
WALLET	72196.04	1901582.00	1829385.96
UPI	47618.63	2197323.00	2149704.37
UNKNOWN	0.00	2301530.00	2301530.00

Recommendations:

- **Promote UPI** through incentives and reduce dependency on COD to lower payment processing costs.
- **Improve payment method data capture** to measure total fee leakage and profitability impact accurately.

REVENUE LEAKAGE BREAKDOWN

INSIGHTS

- **Discounts** were identified as the largest revenue leakage source, contributing significantly more leakage than returns, logistics costs, or payment gateway fees.
- **Logistics and payment gateway fees** contributed comparatively lower but still meaningful operational leakage.

Recommendations:

- Optimizing promotional strategy and reducing unnecessary discounts could provide the highest profitability improvement opportunity.

Total_discount_Leakage	Total_return_leakage	Total_logistics_leakage	Total_payment_gateway_fee_leakage
1654459	520032.00	338224	332751.77

order_id	Total_Revenue	Total_discount_amount	Total_return_amount	Total_logistics_cost	Total_payment_gateway_fee
11487	8975.00	413	0	98	188.48
16156	8960.00	0	0	0	313.60
17067	8960.00	0	0	121	107.52
15622	8955.00	0	0	0	0.00
14303	8950.00	0	0	35	179.00
5840	8950.00	0	0	132	187.95
6737	8950.00	0	0	0	313.25
12842	8945.00	298	0	0	178.90
7612	8930.00	0	0	119	107.16
8514	8905.00	240	0	0	311.68
13784	8900.00	159	0	129	0.00

PRODUCT PROFIT RANKING

INSIGHTS

- **Grocery category** dominated the profit rankings, with 3 of the top 5 products generating the highest net profit contribution despite multiple leakage deductions
- Several products maintained strong profitability even after discounts, returns, logistics, and payment fee impacts, indicating **highly resilient product performance**

Recommendations:

- **Prioritize inventory planning and marketing spend toward top-ranked profitable products** to maximize overall profitability.
- **Improve product and brand categorization** while reviewing low-performing products for repricing or discontinuation.

product_id	category	sub_category	brand	Net_Profit_Contribution	Product_Rank
963	Grocery	SubCat-2	BrandC	32701.02	1
887	Home	SubCat-10	UNKNOWN	31069.19	2
549	Grocery	SubCat-8	UNKNOWN	31039.00	3
1019	Fashion	SubCat-10	BrandB	28230.18	4
124	Grocery	SubCat-3	BrandC	26484.64	5
1677	Beauty	SubCat-8	UNKNOWN	26144.26	6
633	Fashion	SubCat-8	BrandA	25250.94	7
1701	unknown	SubCat-8	BrandB	25175.24	8
1154	Home	SubCat-1	BrandC	25069.81	9
1728	Home	SubCat-11	BrandB	24529.97	10
1219	Beauty	SubCat-6	BrandD	24433.84	11

CATEGORY MARGIN STABILITY

INSIGHTS

- High variation across all categories suggests marketplace-wide pricing and cost management inconsistency rather than isolated category issues.

Recommendations:

- Implement tighter pricing controls and category-specific margin management strategies, especially for Grocery and Electronics categories.

category	Margin_variation
Grocery	134.96
Electronics	120.74
Home	103.19
unknown	102.72
Beauty	102.46
Fashion	95.05

HIGH-RISK CUSTOMERS

INSIGHTS

- Customer return behavior was highly concentrated among a certain group of customers, with several customers returning all completed orders, indicating elevated return-risk patterns and potential profitability impact.

Recommendations:

- Implement customer-level return monitoring, stricter return policy controls, and risk-based intervention strategies to reduce excessive return behavior and operational costs.

customer_id	Customer_Return_Rate
12	100.00
148	100.00
632	100.00
764	100.00
1007	100.00
1078	100.00
1141	100.00
1151	100.00
1299	100.00
1314	100.00
1359	100.00
...	...

EXECUTIVE PROFITABILITY SUMMARY

Total_Sales	Total_Profit	Total_discounts	Total_returns_loss	Total_logistics_cost	Total_payment_fees
19308912.00	10583046.00	1654459	520032.00	338224	332751.77

- Despite multiple operational leakages, the marketplace maintained **strong overall profitability with a healthy final Net Margin.**
- The analysis highlights significant opportunities to improve realized profit through tighter control of **Discounts, Returns, and Operational expenses.**

Total Sales	₹19,308,912	(100%)
Less: Product Cost	₹ 8,725,866	(45.2%)
Gross Profit	₹10,583,046	(54.8%)
Less: Discounts	₹ 1,654,459	(8.6%)
Less: Returns	₹ 520,032	(2.7%)
Less: Logistics	₹ 338,224	(1.8%)
Less: Payment Fees	₹ 332,752	(1.7%)
Net Profit	₹ 7,737,579	(40.1%)

The background features abstract geometric patterns in teal. In the top-left and bottom-left corners, there are nested rectangular outlines. In the top-right and bottom-right corners, there are clusters of small teal circles arranged in a grid-like pattern. A diagonal teal line runs from the top-left towards the bottom-right, intersecting the other elements.

THANK YOU